LEVEL-II (C.W)

PROBLEMS ON CLASSICAL DEFINITION OF PROBABILITY

1. A positive integer is selected at random. If A be the event that it is divisible by 5 and B be the event that it has zero at the units place, then $A \cup \overline{B}$ is

1) An impossible event
2) a certain event
3) $\overline{A} \cap B$
4) the event that the number has a non-zero digits at the units place

2. There are 7 seats in a row. Three persons take seats at random the probability that the middle seat is always occupied and no two persons are consecutive is

1) $\frac{9}{70}$ 2) $\frac{9}{35}$ 3) $\frac{4}{35}$ 4) $\frac{6}{35}$

3. If the letters of the word MISSISSIPPI are arranged at random, the probability that all the 4 S's appear consecutively is

1) $\frac{\angle 8}{\angle 11}$ 2) $\frac{\angle 4}{\angle 11}$ 3) $\frac{\angle 8 \angle 4}{\angle 11}$ 4) $\frac{\angle 6}{\angle 11}$

4. If 10 persons are to sit around a round table, the odds against two specified persons sitting together is

1) $\frac{1}{9}$ 2) $\frac{2}{9}$ 3) 7 to 2 4) 2 to 7

5. The probability that the birthdays of 6 girls will fall on 6 different calender months of a year is

1) $\frac{1}{12^5}$ 2) $\frac{{}^6c_6}{12^6}$ 3) $\frac{{}^{12}p_6}{12^6}$ 4) $\frac{1}{12^{11}}$

6. Out of numbers 1 to 9, two numbers are chosen at random, so that their sum is even number. The probability that the two chosen numbers are odd is

1) $\frac{5}{18}$ 2) $\frac{13}{18}$ 3) $\frac{5}{8}$ 4) $\frac{1}{8}$

7. Three integers are chosen at random without replacement from the 1st 20 integers. The probability that their product is odd is

1) $\frac{3}{19}$ 2) $\frac{2}{19}$ 3) $\frac{1}{19}$ 4) $\frac{4}{19}$

8. There are m persons sitting in a row. Two of them are selected at random. The probability that the two selected persons are together

1) $\frac{{}^{m-1}c_2}{{}^mc_2}$ 2) $\frac{m-1}{{}^mc_2}$ 3) $\frac{m-3}{{}^mc_2}$ 4) $1-\frac{{}^{m-1}c_2}{{}^mc_2}$

9. Consider a lottery that sells n^2 tickets and awards n prizes. If one buys n tickets the probability of his winning is i.e., getting at least one prize is

1) $\frac{n}{{}^{n^2}c_n}$ 2) $1-\frac{{}^{(n^2-n)}c_n}{{}^{n^2}c_n}$ 3) $\frac{(n^2-n)}{n^2}$ 4) $\frac{1}{{}^{n^2}c_n}$

10. Dialling a telephone number to his daughter an old man forgets the last two digits and dialled at random remembering only that they are different. The probability that the number dialled is correct is

1) $\frac{1}{10}$ 2) $\frac{1}{45}$ 3) $\frac{1}{90}$ 4) $\frac{1}{135}$

11. 12 balls are distributed among 3 boxes. The probability that the 1st box contains 3 balls is

1) $\left(\frac{2}{3}\right)^{10}$ 2) $\frac{100}{9\times 3^{10}}$ 3) $\frac{110}{9}\times\left(\frac{2}{3}\right)^{10}$ 4) $\frac{100}{9}$

12. Three houses are available in a locality. Three persons apply for the houses. Each applies for one house without consulting others. The probability that all three apply for the same house is

1) 1/9 2) 2/9 3) 7/9 4) 8/9

13. There are 2 locks on the door and the keys are among the six different ones you carry in your pocket. In a hurry you dropped one somewhere. The probability that you can still open the door is

1) $\frac{1}{2}$ 2) $\frac{1}{3}$ 3) $\frac{2}{3}$ 4) $\frac{1}{4}$

14. Two friends A and B have equal number of sons. There are 3 cinema tickets which are to be distributed among the sons of A and B. The probability that all the tickets go to sons of B is 1/20. The number of sons, each of them having is

1) 2 2) 4 3) 5 4) 3

15. Three groups of children contain respectively 3 girls & 1 boy; 2 girls & 2 boys; 1 girl & 3 boys. One child is selected at random from each group. The chance that the three selected children consists of one girl and 2 boys is

1) $\frac{9}{32}$ 2) $\frac{11}{32}$ 3) $\frac{13}{32}$ 4) $\frac{7}{32}$

16. **A letter is taken out at random from the word ASSISTANT and an other from STATISTICS. The probability that they are the same letters is**

1) $\frac{13}{90}$ 2) $\frac{17}{90}$ 3) $\frac{19}{90}$ 4) $\frac{15}{90}$

17. **If a number x is selected from the 1st 100 natural numbers at random, then the probability that $x+\frac{100}{x}>50$ is**

1) $\frac{9}{20}$ 2) $\frac{1}{2}$ 3) $\frac{11}{20}$ 4) $\frac{5}{20}$

18. **Three newly wedded couples are dancing at a function. If the partner is selected at random the chance that all the husbands are not dancing with their own wives is**

1) $\frac{1}{3}$ 2) $\frac{1}{6}$ 3) $\frac{5}{6}$ 4) $\frac{2}{3}$

19. **2n boys are randomly divided into two subgroups containing n boys each. The probability that the two tallest boys are in different groups is**

1) $\frac{1}{2}$ 2) $\frac{n}{2n-1}$ 3) $\frac{(n-1)}{(2n-1)}$ 4) $\frac{2n}{2n-1}$

20. **A bag contains apples and oranges, five in all and atleast one of each, all combinations being equally likely. If one fruit is selected at random from the bag, assuming all fruits are distinguishable, the probability that it is an orange is**

1) $\frac{1}{20}$ 2) $\frac{1}{10}$ 3) $\frac{1}{5}$ 4) $\frac{1}{2}$

21. **There are 10 stations between A and B. A train is to stop at three of these 10 stations. The probability that no two of these stations are consecutive is**

1) $\frac{^{8}c_3}{^{10}c_3}$ 2) $\frac{^{9}c_3}{^{10}c_3}$ 3) $\frac{^{7}c_3}{^{10}c_3}$ 4) $\frac{^{10}c_3}{^{12}c_3}$

22. **If the papers of 4 students can be checked by any one of the seven teachers then the probability that all the four papers are checked by exactly two teachers is**

1) $\frac{6}{49}$ 2) $\frac{2}{7}$ 3) $\frac{32}{343}$ 4) $\frac{2}{343}$

23. **A box contains 100 tickets numbered 1, 2, ..., 100. Two tickets are chosen at random. It is given that the maximum number on the two chosen tickets is not more than 10. The probability that the minimum number on them is 5 is**

1) $\frac{1}{9}$ 2) $\frac{2}{9}$ 3) $\frac{3}{9}$ 4) $\frac{4}{9}$

24. **Five horses are in a race. Mr. A selects two of the horses at random and bets on them. The probability that Mr. A selected the winning horse is**

1) $\frac{3}{5}$ 2) $\frac{1}{5}$ 3) $\frac{2}{5}$ 4) $\frac{4}{5}$

25. **Two integers x and y are chosen with replacement out of the set $\{0, 1, 2, 3, ..., 10\}$. Then the probability that $|x-y|>5$ is**

1) $\frac{81}{121}$ 2) $\frac{30}{121}$ 3) $\frac{25}{121}$ 4) $\frac{20}{121}$

26. **A determinant is chosen at random from the set of all determinants of order 2 with elements 0 or 1 only. The probability that the determinant is positive is**

1) $\frac{3}{16}$ 2) $\frac{3}{8}$ 3) $\frac{5}{8}$ 4) $\frac{7}{8}$

27. **A team of 8 couples (husband and wife) attend a lucky draw in which 4 persons picked up for a prize. Then the probability that there is atleast one couple is**

1) $^{11}/_{39}$ 2) $^{12}/_{39}$ 3) $^{14}/_{39}$ 4) $^{15}/_{39}$

28. **A subset A of X = {1, 2, 3, ..., 100} is chosen at random. The set X is reconstructed by replacing the elements of A and another subset B of X is chosen at random. The probability that $A\cap B$ contains exactly 10 elements is**

1) $\frac{1}{10}$ 2) $^{100}c_{10}\left(\frac{1}{2}\right)^{100}$ 3) $^{100}c_{10}\left(\frac{1}{4}\right)^{100}$ 4) $^{100}c_{10}\frac{3^{90}}{4^{100}}$

29. **A has 3 tickets of a lottery containing 3 prizes and 9 blanks. B has two tickets of another lottery containing 2 prizes and 6 blanks. The ratio of their chances of success is**

1) $\frac{32}{55}:\frac{15}{28}$ 2) $\frac{32}{55}:\frac{13}{28}$ 3) $\frac{34}{55}:\frac{13}{28}$ 4) $\frac{34}{55}:\frac{15}{28}$

30. S = {1, 2, 3, ... 11} if 3 numbers are chosen at random from S, the probability for they are in G.P.

1) $\frac{7}{^{11}c_3}$ 2) $\frac{9}{^{11}c_3}$ 3) $\frac{5}{^{11}c_3}$ 4) $\frac{4}{^{11}c_3}$

31. If 10 identical apples are distributed among 6 persons at random then the probability that atleast one of them will receive none is

1) $\frac{6}{143}$ 2) $\frac{^{14}c_4}{^{15}c_5}$ 3) $\frac{137}{143}$ 4) $\frac{1}{143}$

32. In a game called 'odd man out', 4 persons toss a coin to decide who will buy refreshments for the entire group. A person who gets an outcome different from that of the rest of the members of the group is called the odd man out. The probability that there is a loser in any game is

1) 1/3 2) 1/4 3) 1/2 4) 1/5

33. Two numbers 'a' and 'b' are chosen at random from the numbers 1,2,3,....30. The chance that $a^2 - b^2$ is divisible by '3' is

1) $\frac{9}{87}$ 2) $\frac{12}{87}$ 3) $\frac{15}{87}$ 4) $\frac{47}{87}$

34. Two numbers are selected at random from 1,2,3,.....,100 and are multiplied, then the probability that the product thus obtained is divisible by 3 correct to two places of decimal is

1) 0.45 2) 0.55 3) 0.25 4) 0.35

35. Four digit numbers with different digits are formed using the digits 1,2,3,4,5,6,7,8. One number from them is picked up at random. The chance that the selected number contains the digit '1' is

1) $\frac{1}{2}$ 2) $\frac{1}{4}$ 3). $\frac{1}{8}$ 4) $\frac{1}{16}$

36. Three numbers are chosen at random from the set {1, 2, 3, 4, 5, 6, 7, 8, 9, 10}. The probability that the smallest of the three numbers chosen is even is

1) $\frac{5}{12}$ 2) $\frac{7}{12}$ 3) $\frac{5}{6}$ 4) $\frac{1}{6}$

37. Out of 5 digits 0, 3, 3, 4, 5 five digit numbers are formed. If one number is selected at random out of them. The probability that it is divisible by 5 is

1) $\frac{3}{16}$ 2) $\frac{5}{16}$ 3) $\frac{7}{16}$ 4) $\frac{9}{16}$

38. From the first 100 natural numbers a number is chosen at random, the probability for it to be a composite number is

1) $\frac{74}{100}$ 2) $\frac{24}{100}$ 3) $\frac{25}{100}$ 4) $\frac{26}{100}$

39. From 101 to 1000 natural numbers a number is taken at random. The probability that the number is divisible by 17 is

1) $\frac{58}{900}$ 2) $\frac{58}{100}$

3) $\frac{53}{900}$ 4) $\frac{53}{1000}$

40. 3 coins are marked with a, b; b,c; c,a. All are tossed, probability that two of the faces shows the same letter is

1) $\frac{3}{8}$ 2) $\frac{1}{8}$ 3) $\frac{3}{4}$ 4) $\frac{1}{4}$

41. A magical die is so loaded that the probability of any face appearing is proportional to the number of points on its face. The probability of an odd number appearing is

1) $\frac{2}{7}$ 2) $\frac{3}{7}$ 3) $\frac{4}{7}$ 4) $\frac{5}{7}$

42. A die is loaded such that 6 turning upwards is twice as often as 1 and three times as any other face. The chance that we get a face with one point when we throw such a die is

1) $\frac{6}{17}$ 2) $\frac{3}{17}$ 3) $\frac{2}{17}$ 4) $\frac{5}{17}$

43. A fair die is thrown untill a face with less than 5 points is obtained. The probability of obtaining not less than 2 points on the last throw is

1) $\frac{1}{4}$ 2) $\frac{1}{2}$ 3) $\frac{3}{4}$ 4) $\frac{1}{8}$

44. An ordinary die has four blank faces. One face marked 2, an other marked 3) Then the probability of obtaining a total of exactly 12 in 5 throws is

1) $\frac{15}{6^4}$ 2) $\frac{10}{6^4}$ 3) $\frac{5}{6^4}$ 4) $\frac{6}{6^4}$

45. A and B throw a symmetrical die each. The odds against ' A' not throwing a number greater than B is

1) 1 to 5 2) 5 to 1 3) 7 to 5 4) 5 to 7

46. Three faces of a fair die are yellow, two faces red and one blue. The die is thrown twice. The probability that 1st throw will give an yellow face and the second a blue face is

1) $\frac{1}{6}$ 2) $\frac{1}{9}$ 3) $\frac{1}{12}$ 4) $\frac{1}{3}$

47. In a single throw with a pair of dice, the total score of occurrence for which the probability is maximum is

1) 5 2) 6 3) 7 4) 8

48. If three dice are thrown, then the probability that they show the numbers in A.P. is

1) $\frac{1}{36}$ 2) $\frac{1}{18}$ 3) $\frac{2}{9}$ 4) $\frac{5}{18}$

49. It is given that there are 52 Thursdays in a leap year. Then the probability that it will have 52 Fridays is

1) $\frac{2}{5}$ 2) $\frac{4}{5}$ 3) $\frac{1}{5}$ 4) $\frac{3}{5}$

50. Two children were picked at random and found to have been born in 1992 then the probability that exactly one of them is born on 29^{th} February is

1) 0 2) $\frac{1}{366\times183}$ 3) $\frac{365}{366\times183}$ 4) $\frac{731}{(366)^2}$

51. Five cards are drawn at random from a well shuffled pack of 52 playing cards. The probability that four of them may have the same face value is

1) $\frac{{}^{13}c_1\times{}^{4}c_1\times{}^{48}c_1}{{}^{52}c_5}$ 2) $\frac{{}^{13}c_1\times{}^{48}c_1}{{}^{52}c_5}$

3) $\frac{{}^{13}c_5}{{}^{52}c_5}$ 4) $\frac{{}^{13}c_4\times{}^{4}c_1}{{}^{52}c_5}$

52. The probability of getting 9 cards of the same suit by a particular hand at a game of bridge is

1) $\frac{{}^{13}c_9}{{}^{52}c_{13}}$ 2) $\frac{{}^{13}c_9\times{}^{39}c_4}{{}^{52}c_{13}}$

3) $\frac{{}^{4}c_1\times{}^{13}c_9\times{}^{39}c_4}{{}^{52}c_{13}}$ 4) $\frac{{}^{4}c_1\times{}^{13}c_9}{{}^{52}c_{13}}$

53. In a hand at which the probability that 4 queens are held by a specified player is

1) $\frac{4\times{}^{48}c_9}{{}^{52}c_{13}}$ 2) $\frac{{}^{48}c_9}{{}^{52}c_{13}}$ 3) $\frac{4\times{}^{48}c_{13}}{{}^{52}c_{13}}$ 4) $\frac{2\times{}^{48}c_{13}}{{}^{52}c_{13}}$

54. Two cards drawn one after another at random without replacement. The probability that both of them may have the same face value is

1) $\frac{1}{221}$ 2) $\frac{1}{169}$ 3) $\frac{1}{17}$ 4) $\frac{1}{19}$

55. Two cards are drawn at random from a pack of 52 cards. The probability of getting atleast a spade and an ace is

1) $\frac{1}{34}$ 2) $\frac{8}{221}$ 3) $\frac{1}{26}$ 4) $\frac{2}{51}$

56. The probability of getting different suit cards and different denomination cards when two cards are drawn from a pack is

1) $\frac{13}{17}$ 2) $\frac{13}{34}$ 3) $\frac{12}{17}$ 4) $\frac{6}{17}$

57. There are 3 cards. Each is painted red on one side and black on the other side. The cards are randomly placed in a row. The probability that no two consecutive cards show red is

1) $\frac{3}{8}$ 2) $\frac{5}{8}$ 3) $\frac{5}{6}$ 4) $\frac{1}{8}$

58. A bag contains 10 white and 3 black balls. Balls are drawn one by one without replacement till all the black balls are drawn. The probability that the test will come to an end at the 7^{th} draw is

1) $\frac{{}^{10}c_4\times{}^{3}c_2}{{}^{13}c_6}\times\frac{1}{7}$ 2) $\frac{{}^{10}c_4\times{}^{3}c_2}{{}^{13}c_6}\times\frac{2}{7}$

3) $\frac{{}^{10}c_4\times{}^{3}c_2}{{}^{13}c_7}$ 4) $\frac{{}^{10}c_5\times{}^{3}c_2}{{}^{13}c_7}$

59. Four tickets numbered 00, 01, 10, 11 are placed in a bag. A ticket is drawn at random and replaced. Again a ticket is drawn at random. The probability that the sum of the numbers on the tickets drawn is 21

1) $\frac{1}{4}$ 2) $\frac{1}{8}$ 3) $\frac{1}{16}$ 4) $\frac{1}{12}$

60. Two squares of a chess board having 8×8 squares are selected at random the probability that they have a side in common is

1) $\frac{56}{{}^{64}c_2}$ 2) $\frac{112}{{}^{64}c_2}$ 3) $\frac{168}{{}^{64}c_2}$ 4) $\frac{268}{{}^{64}c_2}$

61. If 16 squares of unit size are selected at random on a chess board, the probability that they form a square of 4×4 is

1) $\frac{25}{^{64}C_{16}}$ 2) $\frac{25}{^{64}C_{4}}$ 3) $\frac{36}{^{64}C_{16}}$ 4) $\frac{36}{^{64}C_{4}}$

PROBLEMS ON ADDITION THEOREM

62. Three numbers are chosen at random from {1,2,3,.....,10}. The probability that minimum of chosen number is 3 or maximum is 7, is
1) 11/30 2) 11/40 3) 1/7 4) 1/8

63. Out of 15 persons 10 can speak Hindi and 8 can speak English. If two persons are chosen at random, then the probability that one peron speaks Hindi only and the other speaks both Hindi and English is

1) $3/5$ 2) $7/12$ 3) $1/5$ 4) $2/5$

64. An electric bulb will last for 150 days or more with a probability 0.7 and it will last for at the most 160 days with probability 0.8. The probability that the bulb will last between 150 and 160 days is
1) 0.1 2) 0.3 3) 0.5 4) 0.4

PROBLEMS ON CONDITIONAL PROBABILITY

65. If $P(A)=\frac{3}{8}, P(B)=\frac{5}{8}$ & $P(A\cap B)=\frac{1}{4}$, then $P\left(\frac{\bar{A}}{B}\right)=$

1) $\frac{1}{5}$ 2) $\frac{2}{5}$ 3) $\frac{3}{5}$ 4) $\frac{4}{5}$

66. For a biased die, the probability for different faces to turn up are givenbelow:

Face:	1	2	3	4	5	6
Probability:	0.10	0.32	0.21	0.15	0.05	0.17

Such a die is tossed once and you are told that face 1 or 2 has turned up. The probability that it is face 1 is

1) $\frac{16}{21}$ 2) $\frac{5}{21}$ 3) $\frac{4}{7}$ 4) $\frac{3}{7}$

67. Two dice are thrown. The probability of getting a sum of 7 points, if it is known that the two dice are showing different numbers is

1) $\frac{1}{6}$ 2) $\frac{1}{5}$ 3) $\frac{1}{4}$ 4) $\frac{1}{8}$

68. Two symmetrical dice are rolled. If the numbers thrown up on them are different, the probability of getting an even number as the sum of the numbers is

1) $\frac{1}{5}$ 2) $\frac{2}{5}$ 3) $\frac{3}{5}$ 4) $\frac{4}{5}$

69. Three dice are rolled. If no two dice show the same face, the probability that one is an ace

1) $\frac{1}{4}$ 2) $\frac{1}{3}$ 3) $\frac{1}{2}$ 4) $\frac{1}{8}$

70. In selecting 2 cards one at a time with replacement from a deck, the probability that the second card is an ace, given that the 1st card was a face card is

1) $\frac{1}{13}$ 2) $\frac{2}{13}$ 3) $\frac{3}{13}$ 4) $\frac{4}{13}$

71. One ticket is selected randomly from the set of 100 tickets numbered as {00, 01, 02, 03, 04,05,........,98, 99}. E_1 and E_2 denote the sum and product of the digits of the number of the selected ticket. The value of $P\left(\frac{E_1=9}{E_2=0}\right)$ is

1) 1/19 2) 2/19 3) 3/19 4) 1/18

72. A piece of equipment will function when all the three components A, B, C are working. The probability of A failing during one year is 0.15 that of B is 0.05 and that of C is 0.10. The probability that the equipment will fail before the end of the year is
1) 0.72675 2) 0.27325 3) 1 4) 0.95

73. A box contains 10 mangoes out of which 4 are rotten. Two mangoes are taken together. If one of them is found to be good, the probability that the other is rotten is

1) $\frac{5}{13}$ 2) $\frac{7}{13}$ 3) $\frac{8}{13}$ 4) $\frac{9}{13}$

PROBLEMS ON INDEPENDENT EVENTS

74. Let A and B be two events such that $P(\overline{A\cup B})=\frac{1}{6}, P(A\cap B)=\frac{1}{4}$ and $P(\bar{A})=\frac{1}{4}$

Then events A and B are (JEE MAIN 2014)
1) equally likely, but not independent
2) equally likely and mutually exclusive
3) mutually exclusive and independent
4) independent but not equally likely

75. A candidate takes three tests in succession and the probability of passing the first test is p. The probability of passing each succeeding test is p or $\frac{p}{2}$ according as he passes or fails in the preceding one. The candidate is selected if he passes at least two tests. The probability that the candidate is selected is (EAM-2014)

1) $p(2-p)$ 2) $p+p^2+p^3$
3) $p^2(1-p)$ 4) $p^2(2-p)$

76. A coin is biased so that the probability of falling head when tossed is 1/4. If the coin is tossed 5 times the probability of obtaining 2 heads and 3 tails, with heads occuring in succession is

1) $\frac{5\times3^3}{4^5}$ 2) $\frac{3^3}{5^4}$ 3) $2\left(\frac{3}{8}\right)^3$ 4) $\frac{3^3}{4^5}$

77. The probability that a teacher will give a surprise test during any class meeting is 3/5. If a student is absent on two days, then the probability that he will miss at least one test is

1) 9/25 2) 4/25 3) 21/25 4) 13/25

78. A person draws a card from a well shuffled pack of 52 playing cards. Replaces it and shuffles the pack. He continues doing so until he draws a spade. The chance that he fails first two times is

1) $\frac{1}{16}$ 2) $\frac{9}{16}$ 3) $\frac{9}{64}$ 4) $\frac{9}{32}$

79. A symmetrical die is thrown 1st and secondly two symmetrical dice are thrown together. The probability that 1st throw was a face with 6 points upward & the second throw was a sum of 6 points

1) $\frac{1}{36}$ 2) $\frac{5}{36}$ 3) $\frac{5}{216}$ 4) $\frac{1}{216}$

80. A box contains 2 black, 3 white, 4 red balls. One ball is drawn at random and kept aside. From the remaining balls another ball is drawn and kept aside the first. The process is repeated till all the balls are drawn from the box. The probability that balls drawn are in the sequence of 2 black, 3 white, 4 red balls is

1) $\frac{{}^2c_2.{}^3c_3.{}^4c_3}{{}^9c_3}$ 2) $\frac{1}{1260}$ 3) $\frac{1}{180}$ 4) $\frac{1}{630}$

81. Two fair dice are tossed. Let X be the event that the first die shows an even number and Y be the event that the second die shows an odd number. The two events X and Y are

1) mutually exclusive
2) independent and mutually exclusive
3) dependent
4) independent

82. A Six faced fair dice is thrown until 1 comes, then probability that 1 comes in even number of trials is

1) $\frac{3}{11}$ 2) $\frac{5}{11}$ 3) $\frac{3}{4}$ 4) $\frac{7}{9}$

83. A and B alternately throw with a pair of dice. Who ever gets a sum of 7 points 1st will win the game. If A starts the game, the probability of his winning is

1) $\frac{4}{11}$ 2) $\frac{5}{11}$ 3) $\frac{6}{11}$ 4) $\frac{3}{11}$

84. A pair of dice is rolled together till a sum of either 5 or 7 is obtained. Then the probability that 5 comes before 7 is

1) $\frac{1}{5}$ 2) $\frac{2}{5}$ 3) $\frac{3}{5}$ 4) $\frac{4}{5}$

85. A man alternately tosses a coin and throws a die beginning with coin. The probability that he gets a head before he gets 5 or 6 on the die is

1) $\frac{1}{4}$ 2) $\frac{1}{2}$ 3) $\frac{3}{4}$ 4) $\frac{1}{8}$

86. A and B alternately cut a card each from a pack of cards with replacement and pack is shuffled after each cut. If A starts the game and the game is continued till one cuts a spade, the respective probabilities of A and B cutting a spade are

1) $\frac{1}{3},\frac{2}{3}$ 2) $\frac{3}{4},\frac{1}{4}$ 3) $\frac{4}{7},\frac{3}{7}$ 4) $\frac{3}{7},\frac{4}{7}$

87. There are four machines and it is known that exactly two of them are faulty. They are tested one by one in a random order till both the faulty machines are identified. Then the probability that only two tests are needed is

1) $\frac{1}{3}$ 2) $\frac{1}{6}$ 3) $\frac{1}{2}$ 4) $\frac{1}{4}$

88. A box contains 24 identical balls of which 12 are white and 12 are black. The balls are drawn at random from the box one at a time with replacement. The probability that a white ball is drawn for the 4th time on the 7th draw is

1) $\frac{5}{64}$ 2) $\frac{27}{32}$ 3) $\frac{5}{32}$ 4) $\frac{1}{2}$

89. A biased coin with probability P, $(0 < p < 1)$ of heads is tossed unitl a head appear for the first time. If the probability that the number of tosses required is even is $\frac{2}{5}$ then P =

1) $\frac{2}{5}$ 2) $\frac{3}{5}$ 3) $\frac{2}{3}$ 4) $\frac{1}{3}$

90. A bag contains 3 white, 3 black and 2 red balls. One by one 3 balls are drawn without replacing them. For only the 3rd ball to be red the probability is

1) $\frac{1}{28}$ 2) $\frac{3}{28}$ 3) $\frac{5}{28}$ 4) $\frac{7}{28}$

91. An urn contains 2 white and 2 black balls. A ball is drawn at random. If it is white it is not replaced into the urn. Otherwise it is replaced along with ball of the same colour. The process is repeated. The probability that the third ball drawn is black is

1) $\frac{15}{29}$ 2) $\frac{11}{30}$ 3) $\frac{7}{30}$ 4) $\frac{23}{30}$

PROBLEMS ON TOTAL PROBABILITY

92. There are 12 unbiased coins in a bag. Out of them 4 coins have head on both the sides. One coin is selected from the bag at random and tossed. The probability of getting a head is

1) $\frac{1}{2}$ 2) $\frac{1}{3}$ 3) $\frac{2}{3}$ 4) $\frac{1}{4}$

93. H is one of the 6 horses entered for a race and is to be ridden by one of the two jokeys A and B. It is 2 to 1 that A rides H in which case all the horses are likely to win. If B rides H, his chance is trebled. Then the odds against H winning is

1) 4 to 13 2) 13 to 4 3) 13 to 5 4) 13 to 7

94. The probability that in a year of 22nd century chosen at random, there will be 53 Sundays is

1) 3/28 2) 2/28 3) 7/28 4) 5/28

95. There are 2 bags one of which contains 3 black and 4 white balls, while the other contains 4 black and 3 white balls. A die is cast, if face 1 or 3 turns up a ball in taken from the 1st bag and if any other face turns up a ball is taken from the second bag. The probability of choosing a black ball is

1) $\frac{5}{21}$ 2) $\frac{10}{21}$ 3) $\frac{11}{21}$ 4) $\frac{6}{21}$

96. An urn contains 5 white and 7 black balls. A second urn contains 7 white and 8 black balls. One ball is transfered from the 1st urn to the 2nd urn without noticing its colour. A ball is now drawn at random from the 2nd urn. The probability that it is white is

1) $\frac{8}{92}$ 2) $\frac{89}{192}$ 3) $\frac{9}{92}$ 4) $\frac{98}{192}$

97. An urn contains 6 white and 4 black balls. A fair die whose faces are numbered from 1 to 6 is rolled and number of balls equal to that of the number appearing on the die is drawn from the urn at random. The probability that all those are white is

1) $\frac{1}{5}$ 2) $\frac{2}{5}$ 3) $\frac{3}{5}$ 4) $\frac{4}{5}$

98. Urn A contains 6 red and 4 black balls and urn B contains 4 red and 6 black balls. One ball is drawn at random from A and placed in B. Then one ball is drawn at random from B and placed in A. If one ball is now drawn from A then the probability that it is found to be red is

1) $\frac{32}{55}$ 2) $\frac{33}{55}$ 3) $\frac{32}{63}$ 4) $\frac{25}{66}$

99. $\frac{2}{3}$ of students of a class are boys and the rest girls. It is given that the probability of a girl getting 1st class is 0.25 and the same for a boy is 0.28. From that class a student is selected at random. The probability that the student is a 1st class student

1) 0.25 2) 0.26 3) 0.27 4) 0.28

100. A bag contains (2n + 1) coins. It is known that n of these have a head on both the sides, whereas the remaining (n + 1) coins are fair. A coin is picked up at random from the bag and tossed. If the probability that the toss results in a head is 31/42, then value of n is

[EAMCET 2013]

1) 10 2) 8 3) 6 4) 25

PROBLEMS ON BAYE'S THEOREM

101. For k = 1,2,3 the box B_k contains k red balls and (k+1) white balls. Let $P(B_1)=\frac{1}{2}$, $P(B_2)=\frac{1}{3}, P(B_3)=\frac{1}{6}$. A box is selected at random and a ball is drawn from it. If a red ball is drawn then the probability that it has come from box B_2 is [EAMCET 2008]

1) 35/78 2) 14/39 3) 10/13 4) 12/13

102. A letter is known to have come from TATANAGAR or CALCUTTA. On the envelope just two consecutive letters TA are visible. The probability that the letter has come from CALCUTTA is

1) 4/11 2) 7/11 3) 1/22 4) 21/22

103. A man is known to speak the truth 2 out of 3 times. He throws a die and reports that it is 'Five'. The probability that it is actually not a 'Five' is

1) 1/2 2) 2/7 3) 1/7 4) 5/7

104. Out of ten coins, one of the coin is known to have heads on both sides. He takes out one coin at random and tosses it 5 times. If it always falls with head upwards, the probability that it is double-headed coin is

1) $\frac{32}{51}$ 2) $\frac{32}{41}$ 3) $\frac{32}{61}$ 4) $\frac{12}{41}$

ADDITIONAL PROBLEMS

105. If 4 people are chosen at random, then the probability that no two of them were born on the same day of the week is

1) $\frac{^7P_4}{7^4}$ 2) $\frac{^7C_4}{7^4}$ 3) $\frac{7!}{7^4}$ 4) $\frac{1}{7^4}$

106. What is the probability that the birthdays of six people will fall in exactly two calender months

1) $\frac{^{12}C_2.(2^6-2)}{(12)^6}$ 2) $\frac{^{12}C_2.(2^6+2)}{(12)^6}$

3) $\frac{^{12}C_2.(2^6-4)}{(12)^6}$ 4) $\frac{^{12}C_2.(2^6)}{(12)^6}$

107. The odds that book be reviewed favourably by three independent critics are 5 to 2, 4 to 3 and 3 to 4 respectively. The probability that of the three reviews a majority will be favourable is

1) $\frac{209}{343}$ 2) $\frac{135}{343}$ 3) $\frac{60}{343}$ 4) $\frac{120}{343}$

108. Set S has 4 elements, A and B are subsets of S. The probability that A and B are not disjoint is

1) $\frac{175}{256}$ 2) $\frac{173}{256}$ 3) $\frac{85}{128}$ 4) $\frac{45}{64}$

109. The sum of two natural numbers is 20. Find the chance that their product less than 50 is

1) $\frac{4}{19}$ 2) $\frac{3}{19}$ 3) $\frac{2}{19}$ 4) $\frac{1}{19}$

110. In constructing problem on vectors, the three components of a vector are randomly chosen from the digit 0 to 5 with replacement. The probability that the magnitude of the vector is 5 is

1) $\frac{1}{24}$ 2) $\frac{1}{12}$ 3) $\frac{1}{6}$ 4) $\frac{1}{30}$

111. Three newly wedded couples are dancing at a function. If the partner is selected at random the chance that at least one husband is not dancing with his own wife is

1) $\frac{1}{3}$ 2) $\frac{1}{6}$ 3) $\frac{5}{6}$ 4) $\frac{2}{3}$

112. Team A plays with 5 other teams exactly once. Assuming that for each match the probabilities of a win, draw and loss are equal, then

1) the probability that A wins and loses equal number of matches is $\frac{34}{81}$

2) the probability that A wins and loses equal number of matches is $\frac{17}{81}$

3) the probability that A wins more number of matches than it loses is $\frac{17}{81}$

4) the probability that A loses more number of matches than it wins is $\frac{16}{81}$

113. Two fair dice are rolled simultaneously. It is found that one of them shows odd prime number. The probability that remaining die also shows an odd prime number, is equal to

1) 1/5 2) 2/5 3) 3/5 4) 4/5

114. Thirty-two players ranked 1 to 32 are playing in a knockout tournament. Assume that in every match between any two players, the better-ranked player wins, the probability that ranked 1 and ranked 2 players are winner and runner up respectively is

1) 16/31 2) 1/2 3) 17/31 4) 18/31

LEVEL-II (C.W.) - KEY

1) 2	2) 3	3) 3	4) 3	5) 3	6) 3	7) 2
8) 2	9) 2	10) 3	11) 3	12) 1	13) 3	14) 4
15) 3	16) 3	17) 3	18) 3	19) 2	20) 4	21) 1
22) 1	23) 1	24) 3	25) 2	26) 1	27) 4	28) 4
29) 3	30) 4	31) 3	32) 3	33) 4	34) 2	35) 1
36) 1	37) 3	38) 1	39) 3	40) 3	41) 2	42) 2
43) 3	44) 3	45) 4	46) 3	47) 3	48) 2	49) 2
50) 3	51) 2	52) 3	53) 2	54) 3	55) 3	56) 3
57) 2	58) 1	59) 2	60) 2	61) 1	62) 2	63) 3
64) 3	65) 3	66) 2	67) 2	68) 2	69) 3	70) 1
71) 2	72) 2	73) 3	74) 4	75) 4	76) 3	77) 3
78) 2	79) 3	80) 2	81) 4	82) 2	83) 3	84) 2
85) 3	86) 3	87) 1	88) 3	89) 4	90) 3	91) 4
92) 3	93) 3	94) 4	95) 3	96) 2	97) 1	98) 1
99) 3	100) 1	101) 2	102) 1	103) 4	104) 2	105) 1
106) 1	107) 1	108) 1	109) 1	110) 1	111) 3	112) 2
113) 1	114) 1					

LEVEL-II (C.W.) - HINTS

1. A-unit place contains 5, 0; B-unit place contains 0

   $A \cup \overline{B} = S$

2. .n (S) = $^7p_3 = 210$

   n (E) = $^2c_1 {}^2c_1 {}^1c_1 3!$, $P(E) = \dfrac{4}{35}$

3. $n(S) = \dfrac{11!}{4!4!2!}$ $\quad n(E) = \dfrac{8!}{4!\,2!}$

4. $n(S) = 9!$; $n(E) = 8! \times 2!$

   $P(E) = \dfrac{2}{9}$ $\quad P(\overline{E}) = 1 - \dfrac{2}{9} = \dfrac{7}{9}$

   $P(\overline{E}) : P(E) = \dfrac{7}{9} : \dfrac{2}{9} = 7:2$

5. $n(S) = 12^6$; $n(E) = {}^{12}P_6$

6. Req. Prob. $= \dfrac{^5c_2}{^5c_2 + {}^4c_2}$

7. $n(S) = {}^{20}c_3$ $\quad n(E) = {}^{10}c_3$

8. $n(S) = {}^mc_2$ $\quad n(E) = m-1$

9. Req. Prob. $= 1 - \left(\dfrac{^{(n^2-n)}c_n}{^{n^2}c_2}\right)$

10. $n(S) = {}^{10}P_2$ $\quad n(E) = 1$

11. $n(S) = 3^{12}$ $\quad n(E) = {}^{12}c_3 \,.\, 2^9$

12. Let the houses be A, B, C.

    Each one can apply for any one of these. The number of ways they can apply is $3^3 = 27$

    All three may opt for A or B or C ; $p = \dfrac{3}{27} = \dfrac{1}{9}$

13. $n(S) = {}^6c_2$ $\quad n(E) = {}^5c_2$

14. $\dfrac{^nc_3}{^{2n}c_3} = \dfrac{1}{20} \Rightarrow n = 3$

15. P(BBG) + P(BGB) + P(GBB) =

    $\dfrac{1}{4}.\dfrac{2}{4}.\dfrac{1}{4} + \dfrac{1}{4}.\dfrac{2}{4}.\dfrac{3}{4} + \dfrac{3}{4}.\dfrac{2}{4}.\dfrac{3}{4} = \dfrac{13}{32}$

16. $P(A) + P(S) + P(T) + P(I)$

    $= \dfrac{2}{9} \times \dfrac{1}{10} + \dfrac{3}{9} \times \dfrac{3}{10} + \dfrac{2}{9} \times \dfrac{3}{10} + \dfrac{1}{9} \times \dfrac{2}{10}$

17. $n(S) = {}^{100}C_1, n(A) = {}^{55}C_1$

    {1, 2, 50, 51, ... 100, 49, 48}

18. P(all the husbands not dancing with their own wives) = 1- P(all husbands are dancing with their own wives) $= 1 - \dfrac{1}{6} = \dfrac{5}{6}$

19. $n(S) = \dfrac{2n!}{n!\,n!}$; $n(A) = \dfrac{(2n-2)!\,2!}{(n-1)!\,(n-1)!}$

20. Bag may contain (A, 0) = (1, 4) (4, 3) (3, 2) (4, 1)

    Req. Prob $\dfrac{1}{4}\left[\dfrac{4}{5} + \dfrac{3}{5} + \dfrac{2}{5} + \dfrac{1}{5}\right] = \dfrac{1}{2}$

21. Req. Prob $= \dfrac{^{(n-r+1)}c_r}{^nc_r} = \dfrac{^8c_3}{^{10}c_3}$

22. $n(S) = 7.7.7.7 = 7^4$

    $n(A) = 7_{C_2}\left(2^4 - 2\right) = 21 \times 14$ $\qquad P(A) = \dfrac{6}{49}$

23. $n(S) = {}^{10}c_2 = 45$
$n(A) = 5$ {(5, 6) (5, 7) (5, 8) (5, 9) (5, 10)}

24. Req. Prob. $= 1-\frac{{}^4c_2}{{}^5c_2} = \frac{2}{5}$

25. $n(S) = 11\times 11$; $n(E) = 30$; $n(E) = 15 + 15$

26. $n(E) = 3$; $n(S) = 16$

27. $R.P = 1-\left(\frac{16}{16}\right)\left(\frac{14}{15}\right)\left(\frac{12}{14}\right)\left(\frac{10}{13}\right)$

28. $n(S) = 4^{10}$
$n(A) = 100_{C_{10}}.(3)^{90}$

29. Success means getting atleast one prize.
$p(A) = 1-\frac{{}^9C_3}{{}^{12}C_3} : p(B) = 1-\frac{{}^6C_2}{{}^8C_2}$

30. Favourable cases (1,2,4), (2,4,8), (1,3,9), (4,6,9)
Required probability $= \frac{4}{11_{C_3}}$

31. Req. = 1 – prob. of each receiving atleast one
$= 1-\frac{{}^{10-1}c_{6-1}}{{}^{10+6-1}c_{6-1}}$

32. $n(S) = 2^4 = 16$
$n(A) = {}^4c_1 + {}^4c_1 = 8$, $p(A) = \frac{8}{16} = \frac{1}{2}$

33. $a^2 - b^2$ is divisible by '3' if either both a and b are divisible by 3 or neither of them is divisible by '3'
$n(S) = {}^{30}C_2$, $n(E) = {}^{10}C_2$, $P(E) = \frac{47}{87}$

34. $n(S) = {}^{100}C_2, n(E) = {}^{33}C_1 \times {}^{67}C_1 + {}^{33}C_2$;
$P(E) = \frac{913}{1650} = \frac{83}{150} = 0.55$
The product of two numbers is divisible by '3' if atleast one of the numbers is divisible by 3

35. $P(\bar{E}) = \frac{{}^7p_4}{{}^8p_4} = \frac{7.6.5.4}{8.7.6.4} = \frac{1}{2} \Rightarrow P(E) = \frac{1}{2}$

36. $n(S) = {}^{10}c_3 \quad n(E) = {}^8c_2 + {}^6c_2 + {}^4c_2 + {}^2c_2 = 50$

37. $n(S) = 4\times\frac{4!}{2!} = 48$; $n(A) = \frac{4!}{2!} + \frac{4!-3!}{2!} = 12 + 9 = 21$
$P(A) = \frac{21}{48} = \frac{7}{16}$

38. $n(S) = 100 \quad n(E) = 74$

39. $n(S) = 900 \quad n(E) = 53$

40. $R.P = \frac{6}{8}$

41. $P(E_k)\ \alpha\ E_k$; $P(E_1)+P(E_2)+....+P(E_6) = 1$
$k = \frac{1}{21}$; $P(E_1+E_3+E_5) = \frac{9}{21} = \frac{3}{7}$

42. $P(6) = 2\,P(1)$;
$P(6) = 3\,P(2) = 3\,P(3) = 3\,P(4) = 3\,P(5)$
$P(1)+P(2)+P(3)+P(4)+P(5)+P(6) = 1$
$\therefore P(1) = \frac{3}{17}$

43. Req. Probability $= \frac{P(2\le x\le 4)}{P(x<5)} = \frac{3}{4}$

44. (3 3 3 3) (2 2 2 3 3)
Req. probability $= 5\times\left(\frac{1}{6}\right)^4\left(\frac{4}{6}\right)+10\left(\frac{1}{6}\right)^5 = \frac{30}{6^5} = \frac{5}{6^4}$

45. 'A' not throwing a no $>$ B $\Rightarrow$ 'A' throwing a no $\le$ B
$\therefore P(A) = \frac{1}{6}\left(\frac{1}{6}+\frac{2}{6}+\frac{3}{6}+\frac{4}{6}+\frac{5}{6}+\frac{6}{6}\right) = \frac{21}{36} = \frac{7}{12}$
Odds against A is 5 to 7

46. $P(Y\cap B) = \frac{3}{6}\times\frac{1}{6} = \frac{1}{12}$

47. $n(S) = 6^2 \quad n(E) = \{(1,6)(2,5)(3,4)(4,3)(5,2)(6,1)\}$

48. (1, 2, 3) : (2, 3, 4): (3, 4, 5): (4, 5, 6): (1, 3, 5): (2, 4, 6), Required probability
$= \frac{12}{6^3} \quad n(A) = 6+6, n(s) = 6^3$

49. $P\left(\frac{A}{B}\right) = \frac{4}{5}$

50. $n(S) = (366)^2$; $n(E) = 2(365)$; $P(E) = \frac{365}{183(366)}$

51. $P(E) = \frac{{}^{13}c_1\times{}^{48}c_1}{{}^{52}c_5}$

52. $P(E) = \frac{{}^4c_1\times{}^{13}c_9\times{}^{39}c_4}{{}^{52}c_{13}}$

53. $P(E) = \frac{{}^{48}c_9}{{}^{52}c_{13}}$

54. $P(E)=1-\frac{16}{17}=\frac{1}{17}$

55. n (S) = $^{52}c_{2;}$ n(E) = $^{13}c_1)\ ^3c_1 + {}^{12}c_1)\ ^1c_1 = 51$

$P(E)=\frac{1}{26}$

56. $n(S)={}^{52}c_2\ n(E)={}^4c_2\times13\times12$

57. $R.P=1-\frac{3}{8}$

58. $R.P=\frac{10_{c_4}\times3_{c_2}\times1}{13_{c_6}\times7}$

59. (10, 11) (11, 10); Req. Prob = $2\times\frac{1}{4}\times\frac{1}{4}=\frac{1}{8}$

60. Both the squares selected must lie either in a row or column.

61. $n(S)={}^{64}c_{16}$, n(A) = 25

62. A $\rightarrow$ minimum is 3, B $\rightarrow$ maximum is 7,

$P(A\cup B)=P(A)+P(B)-P(A\cap B)$

$=\frac{{}^7C_2+{}^6C_2-{}^3C_1}{{}^{10}C_3}=\frac{11}{40}$

63. $R.P=\frac{7_{c_1}\times3_{c_1}}{15_{c_2}}$

64. $P(A)=P(\geq150)=0.7$; $P(B)=P(\leq160)=0.8$

$P(A\cap B)=P(A)+P(B)-P(A\cup B)=0.5$

65. $P(A)=\frac{3}{8},\ P(B)=\frac{5}{8},\ P(A\cap B)=\frac{1}{4}$

then $P\left(\frac{\overline{A}}{B}\right)=\frac{P(B\cap\overline{A})}{P(B)}=\frac{P(B)-P(A\cap B)}{P(B)}$

$=1-\frac{P(A\cap B)}{P(B)}=1-\frac{2}{5}=\frac{3}{5}$

66. $P\left(\frac{A}{A\cup B}\right)=\frac{0.10}{0.10+0.32}=\frac{10}{42}=\frac{5}{21}$

67. $P\left(\frac{\text{sum 7pts}}{\text{shows diff no's}}\right)=\frac{6}{6\times5}=\frac{1}{5}$

68. To get sum 2 or 4 or 6 or 8 or 10 or 12

Req. probability $=\frac{12}{30}=\frac{2}{5}$

69. $p\left(\frac{\text{an ace}}{\text{different no}}\right)=\frac{\frac{3\times1\times5\times4}{216}}{\frac{6\times5\times4}{216}}=\frac{1}{2}$

70. Req. Prob = $\frac{4}{52}=\frac{1}{13}$

71. n(E_2) = 19, $n(E_1\cap E_2)=2$; $P\left(\frac{E_1}{E_2}\right)=\frac{2}{19}$

72. $P(\overline{A})=0.15, P(\overline{B})=0.05, P(\overline{C})=0.10$

$P(\overline{A\cap B\cap C})=1-P(A\cap B\cap C)$

= 1 – P(A) P(B) P(C)

73. $\frac{{}^6c_1\,.\,{}^4c_1}{{}^6c_1\,.\,{}^4c_1+{}^6c_2}$

74. $P(A\cup B)=1-\frac{1}{6}=\frac{5}{6},\ P(A)=1-\frac{1}{4}=\frac{3}{4}$

P(A) + P(B) = $P(A\cup B)+P(A\cap B)$

$=\frac{5}{6}+\frac{1}{4}=\frac{13}{12}$ $\quad\therefore P(B)=\frac{13}{12}-\frac{3}{4}=\frac{1}{3}$

$P(A)\neq P(B)\Rightarrow$ A, B are not equally likely

P(A) P(B) = $\frac{3}{4}.\frac{1}{3}=\frac{1}{4}=P(A\cap B)$

$\Rightarrow$ A, B are independent.

75. Let A_1, A_2, A_3 be the events that the candidate passes first test, second test, third test respectively.
The probability that the candidate is selected =

$P\left(A_1A_2A_3\cup A_1A_2\overline{A_3}\cup A_1\overline{A_2}A_3\cup\overline{A_1}A_2A_3\right)$

$=p^3+p^2(1-p)+p(1-p)p/2+(1-p)\frac{p}{2}p$

76. Required probability $=4\times\left(\frac{1}{4}\right)^2\left(\frac{3}{4}\right)^3=2\left(\frac{3}{8}\right)^3$

77. R.P. is $1-[P(E)]^2=1-\left(\frac{2}{5}\right)^2=\frac{21}{25}$

78. P(E) = qqp + qqqp + ...

$=qqp\left[\frac{1}{1-q}\right]=q^2=\left(\frac{3}{4}\right)^2=\frac{9}{16}$

79. Req. prob = $\frac{1}{6}\times\frac{5}{36}=\frac{5}{216}$

80. Req. Prob= $\frac{2}{9}\times\frac{1}{8}\times\frac{3}{7}\times\frac{2}{6}\times\frac{1}{5}\times\frac{4}{4}\times\frac{3}{3}\times\frac{2}{2}\times\frac{1}{1}=\frac{1}{1260}$

81. $P(X)=\frac{3}{6}=\frac{1}{2},\ P(Y)=\frac{1}{2}$

$P(X\cap Y)=\frac{9}{36}=\frac{1}{4}=\frac{1}{2}.\frac{1}{2}=P(X).P(Y)$

82. $R.P=\frac{5}{6}\left(\frac{1}{6}\right)+\left(\frac{5}{6}\right)^3\left(\frac{1}{6}\right)+---$

83. $p=\frac{1}{6},q=\frac{5}{6}$ $\quad p(A)=\frac{p}{1-q^2}$

84. $P(5):P(7)=2:3$ $\quad$ Req. Prob = $\frac{2}{5}$

85. Req. Prob = $\frac{1}{2}+\frac{1}{2}.\frac{2}{3}.\frac{1}{2}+...=\frac{\frac{1}{2}}{1-\frac{2}{3}\frac{1}{2}}=\frac{3}{4}$

86. $p=\frac{1}{4},q=\frac{3}{4}$ $\ P(A)=\frac{p}{1-q^2}=\frac{4}{7}$ $\ P(B)=1-\frac{4}{7}=\frac{3}{7}$

87. $P(A)=\frac{2}{4}.\frac{1}{3}+\frac{2}{4}.\frac{1}{3}=\frac{1}{3}$

88. First 6 drawings must contain 3 white balls and 7th draw must be a white ball

Required prob. = ${}^6c_3\left(\frac{1}{2}\right)^3.\left(\frac{1}{2}\right)^3.\frac{1}{2}$

89. Req. prob. = $qp\left(1+q^2+q^4+.....\right)=\frac{2}{5}$, after then verification

90. (W W R) (W B R) (B B R) (B W R)

Req. Prob=$\frac{3}{8}.\frac{2}{7}.\frac{2}{6}+\frac{3}{8}.\frac{2}{7}.\frac{2}{6}+\frac{3}{8}.\frac{3}{7}.\frac{2}{6}+\frac{3}{8}.\frac{3}{7}.\frac{2}{6}$

91. No. of ways W, W, B; W, B, B, B, W, B; B, B, B

92. $P(E)=\frac{16}{24}=\frac{2}{3}$

93. $P(E)=P(A).P\left(\frac{H}{A}\right)+P(B).P\left(\frac{H}{B}\right)$

94. $p(S_{53})=p(L).p(S_{53}/L)+p(NL).p(S_{53}/NL)$

$=\frac{1}{4}.\frac{2}{7}+\frac{3}{4}.\frac{1}{7}=\frac{5}{28}$

95. $\frac{1}{3}\times\frac{3}{7}+\frac{2}{3}\times\frac{4}{7}=\frac{11}{21}$

96. R.P is $\frac{5}{12}\times\frac{8}{16}+\frac{7}{12}\times\frac{7}{16}$

97. Req. Prob = $\frac{1}{6}\left[\frac{{}^6c_1}{{}^{10}c_1}+\frac{{}^6c_2}{{}^{10}c_2}+....+\frac{{}^6c_6}{{}^{10}c_6}\right]$

98. Required probability =

$\frac{6}{10}.\frac{5}{11}.\frac{6}{10}+\frac{4}{10}.\frac{7}{11}.\frac{6}{10}+\frac{6}{10}.\frac{6}{11}.\frac{5}{10}+\frac{4}{10}.\frac{4}{11}.\frac{7}{10}$

99. $P(E)=P(B).P\left(\frac{E}{B}\right)+P(G).P\left(\frac{E}{G}\right)$

100. $\frac{n}{(2n+1)}(1)+\frac{(n+1)}{(2n+1)}.\frac{1}{2}=\frac{31}{42}\Rightarrow n=10$

101. Req.probability =

$$\frac{(1/3)(2/5)}{(1/2)(1/3)+(1/3)(2/5)+(1/6)(3/7)}$$

102. Req. prob. = $\frac{\frac{1}{7}}{\frac{2}{8}+\frac{1}{7}}$

103. Req. prob. = $\frac{\frac{1}{3}\left(\frac{5}{6}\right)}{\frac{2}{3}\left(\frac{1}{6}\right)+\frac{1}{3}\left(\frac{5}{6}\right)}$

104. A $\rightarrow$ all 5 times Head (special or ordinary coin), B $\rightarrow$ double - headed coin

$$P\left(\frac{B}{A}\right)=\frac{P(A\cap B)}{P(A)}=\frac{\frac{1}{10}}{\frac{1}{10}.1+\frac{9}{10}\left(\frac{1}{2}\right)^5}=\frac{1}{1+\frac{9}{32}}=\frac{32}{41}$$

105. Since a person can be born on any day of a week, he can born in 7 ways. By product rule,
The total number of ways that 4 people can born $=7\times7\times7\times7=7^4$; $\quad\therefore\ n(S)=7^4$
Let A be the event that no two of the 4 people chosen were born on the same day of the week.
Here $n(A)$ = number of ways to arrange the 7 days of the week taken 4 at a time when repetitions are not allowed = ${}^7P_4$ $\quad\therefore\quad P(A)=\frac{{}^7P_4}{7^4}$

106. $P(E)=\frac{{}^{12}C_2.\left(2^6-2\right)}{(12)^6}$

107. $P_1=\frac{5}{7}, P_2=\frac{4}{7}, P_3=\frac{3}{7}$

P(E) = $P_1P_2\overline{P_3}+P_1\overline{P_2}P_3+\overline{P_1}P_2P_3+P_1P_2P_3$

$=\frac{5.4.4+5.3.3+2.4.3+5.4.3}{7.7.7}$

$=\frac{80+45+24+60}{343}=\frac{209}{343}$

108. P(E) = $1-\left(\frac{3}{4}\right)^4$

109. n(E)=4, n(S)=19.

110. $n(S)=6^3=216$; $n(E)=\lfloor 3+\frac{\lfloor 3}{\lfloor 2}=9$

(0, 0, 5) (0, 3, 4)

111. Req. Prob. = $1-\frac{1}{3!}$

112. Probability of equal number of W and L is

$(0)W,(0)L+(1)W,(1)L+(2)W,(2)L$

$=\left(\frac{1}{3}\right)^5+{}^5C_1.{}^4C_1\left(\frac{1}{3}\right)^5+{}^5C_2.{}^3C_2\left(\frac{1}{3}\right)^5=\frac{17}{81}$

113. Odd prime on a dice are 3 and 5.

Let event A=one of the dice shows odd prime and event B= remaining dice also shows odd prime

114. For ranked 1 and 2 players to be winners and runners up res., they should not be paired with each other in any round. Therefore, the required probability$=\frac{30}{31}\times\frac{14}{15}\times\frac{6}{7}\times\frac{2}{3}=\frac{16}{31}$

LEVEL-II (H.W)

1. E, $\overline{E}$ **are mutually exclusive and exhaustive events in a sample space 'S' then maximum value of P(E).** P($\overline{E}$) **is**

1) 1/2 2) 1/3
3) 1/4 4) 1/5

2. In a department there are two Professors, four Readers and 6 Lecturers. A committee of three persons is to be formed out of the staff of the department. The probability that the committee consists of at least two lecturers is

1) $\frac{1}{4}$ 2) $\frac{1}{3}$ 3) $\frac{1}{2}$ 4) $\frac{1}{8}$

3. 10 gentlemen and 6 ladies are to sit for a dinner at a round table. The probability that no two ladies sit together is

1) $\frac{\angle 9\times\angle 5}{\angle 15}$ 2) $\frac{\angle 9\times{}^{10}p_6}{\angle 15}$ 3) $\frac{\angle 9\times{}^{10}c_6}{\angle 15}$ 4) $\frac{\angle 9\times{}^{10}c_5}{\angle 15}$

4. 100 tickets are numbered as 00, 01, 02,... 09, 10, 11, 12, ...,99 out of them one ticket is drawn at random. The probability that the sum of the digits of the number on the ticket is 9 is

1) $\frac{7}{100}$ 2) $\frac{9}{100}$ 3) $\frac{1}{10}$ 4) $\frac{1}{100}$

5. If 5 biscuits are distributed among 3 beggers, the chance that a particular beggar will get 2 biscuits is

1) 80/243 2) 30/125 3) 2/15 4) 3/10

6. From a box containing 10 cards numbered 1 to 10, four cards are drawn together. The probability that their sum is odd is

1) $\frac{1}{2}$ 2) $\frac{11}{21}$ 3) $\frac{10}{21}$ 4) $\frac{1}{21}$

7. From 80 cards numbered 1 to 80, two cards are drawn at random. The probability that the cards have the numbers divisible by 4 is

1) $\frac{15}{316}$ 2) $\frac{17}{316}$ 3) $\frac{19}{316}$ 4) $\frac{21}{316}$

8. Three persons A, B and C are to speak at a function with 5 other persons. If the persons speak in random order, the probability that A speaks before B and C speaks before A in that order is

1) $\frac{3}{\angle 8}$ 2) $\frac{{}^8c_3}{\angle 8}$ 3) $\frac{1}{\angle 3}$ 4) $\frac{1}{\angle 8}$

9. Out of the first 25 natural numbers 2 are chosen at random. The probability for one of the numbers is to be a multiple of 3 and the other to be a multiple of 5 is

1) $\frac{13}{100}$ 2) $\frac{1}{5}$ 3) $\frac{2}{15}$ 4) $\frac{1}{15}$

10. If the letters of the word FLOWER are arranged at random, the probability that the order of the consonants is not changed is

1) $\frac{50}{720}$ 2) $\frac{40}{270}$ 3) $\frac{30}{720}$ 4) $\frac{20}{720}$

11. 7 balls are thrown into 4 bags numbered serially 1, 2, 3 & 4 Then the probability that none of them found in bag number 2 is

1) $\frac{3}{4}$ 2) $\frac{1}{4}$ 3) $\left(\frac{3}{4}\right)^7$ 4) $1-\left(\frac{1}{4}\right)^7$

12. Eight persons numbered 1, 2, 3, ..., 8 to be seated round a circular table at random. The probability that the person numbered 1 sits between 2 and 3 is

1) $\frac{1}{21}$ 2) $\frac{1}{42}$ 3) $\frac{1}{56}$ 4) $\frac{1}{2}$

13. Twenty persons among whom A and B, sit at random around a round table, then the probability that there are any 6 persons between A and B is

1) $\frac{2}{19}$ 2) $\frac{17}{19}$ 3) $\frac{{}^{18}c_6 \times 2!}{19!}$ 4) $\frac{{}^{18}c_6 \times 2! \times 12!}{19!}$

14. If four letters are placed into 4 addressed envelopes at random, the probability that exactly two letters will go wrong is

1) $\frac{1}{2}$ 2) $\frac{1}{3}$ 3) $\frac{1}{4}$ 4) 0

15. In the quadratic equation $ax^2+bx+c=0$, the coefficients a, b, c take distinct values from the set {1, 2, 3}. The probability that the roots of the equation are real is

1) $\frac{2}{3}$ 2) $\frac{1}{3}$ 3) $\frac{1}{4}$ 4) $\frac{2}{5}$

16. Let A = { 1,3,5,7,9 } and B= { 2,4,6,8 }. An element (a,b) of their cartesian product A$\times$ B is chosen at random. The probability that a + b = 9 is

1) $\frac{1}{5}$ 2) $\frac{2}{3}$ 3) $\frac{2}{5}$ 4) $\frac{1}{3}$

17. A set P contains n elements. A function from P to P is picked up at random. The probability that this function is onto is

1) $\frac{\angle n}{n^n}$ 2) $\frac{\angle n-1}{n^n}$ 3) $\frac{\left(n^n-\angle n\right)}{n^n}$ 4) $\frac{\angle n-1}{\angle n}$

18. 20 pairs of shoes are there in a closet. Four shoes are selected at random. The probability that they are pairs is

1) $\frac{{}^{20}c_4}{{}^{40}c_4}$ 2) $\frac{{}^{20}c_2}{{}^{40}c_4}$ 3) $\frac{{}^{20}c_2}{{}^{20}c_2}$ 4) $\frac{{}^{20}c_4}{{}^{20}c_4}$

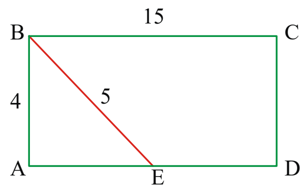
19. From a heap containing 10 pairs of shoes, 8 shoes are selected at random. The probability that 4 correct pairs in the selected shoes is

1) $\frac{{}^{10}C_4}{{}^{20}C_8}$ 2) $\frac{{}^{10}C_4}{{}^{20}C_4}$ 3) $\frac{{}^{10}C_6 \times {}^{8}C_2}{{}^{20}C_8}$ 4) $\frac{{}^{10}C_2}{{}^{20}C_4}$

20. Three different numbers are selected at random from the set A = {1,2,3,....,10}. The probability that the product of two of the numbers is equal to the third is

1) $\frac{3}{4}$ 2) $\frac{1}{40}$ 3) $\frac{1}{8}$ 4) $\frac{1}{4}$

21. In the figure rectangle ABCD, what is the probability that a randomly chosen point that in ABCD lies inside the triangle ABE is



1) $\frac{1}{15}$ 2) $\frac{1}{10}$ 3) $\frac{1}{5}$ 4) $\frac{1}{3}$

22. A bag contains 12 two rupee coins, 7 one rupee coins and 4 half a rupee coins. If three coins are selected at random, p= probability that the sum of the three coins is maximum, q= probability that the sum of the three coins is minimum, r=probability that each coin is of different value. Arrange p,q and r in increasing order of magnitude.

1) p,q,r 2)q,p,r 3)r,q,p 4)r,p,q

23. A and B are to throw 2 dice. If A throws a sum of 9 points, then B's chance of throwing a higher sum is

1) $\frac{1}{2}$ 2) $\frac{1}{3}$ 3) $\frac{1}{6}$ 4) $\frac{5}{9}$

24. A six faced die is so biased that it is twice as likely to show an even number as an odd number when thrown. It is thrown twice. The probability that the sum of the numbers thrown is even is

1) $\frac{1}{3}$ 2) $\frac{4}{9}$ 3) $\frac{5}{9}$ 4) $\frac{6}{9}$

25. The chance of throwing a sum of 6 points with 4 dice is

1) $\frac{6}{6^4}$ 2) $\frac{10}{6^4}$ 3) $\frac{15}{6^4}$ 4) $\frac{5}{6^4}$

26. Two symmetrical dice are rolled once. The probability that both the dice will show 4 is p. The probability for the sum is 8 is q. Then p:q is

1) 4:5 2) 1:5 3) 5:1 4) 5:4

27. It is given that there are 53 Fridays in a leap year. Then the probability that it will have 53 Thursday is

1) $\frac{2}{7}$ 2) $\frac{4}{7}$ 3) $\frac{1}{2}$ 4) $\frac{1}{7}$

28. In a game of bridge the probability of a particular player having only one ace is

1) $\frac{{}^{4}c_1}{{}^{52}c_{13}}$ 2) $\frac{{}^{4}c_1 \times {}^{48}c_{12}}{{}^{52}c_{13}}$

3) $\frac{{}^{48}c_{12}}{{}^{52}c_{13}}$ 4) $\frac{{}^{52}c_{12}}{{}^{52}c_{13}}$

29. In a game of bridge, the player A has received two aces. The probability that his partner has been dealt with the other two aces is

1) $\frac{{}^{2}c_2 \times {}^{48}c_{11}}{{}^{52}c_{13}}$ 2) $\frac{{}^{2}c_2 \times {}^{37}c_{11}}{{}^{39}c_{13}}$

3) $\frac{{}^{2}c_2 \times {}^{37}c_{11}}{{}^{52}c_{13}}$ 4) $\frac{{}^{2}c_2 \times {}^{48}c_{11}}{{}^{39}c_{13}}$

30. 5 cards are drawn at random from a well shuffled pack of 52 playing cards. If it is known that there will be at least 3 hearts, the probability that all the 5 are hearts is

1) $\frac{{}^{13}c_5}{{}^{52}c_5}$

2) $\frac{{}^{13}c_5}{{}^{13}c_3 \times {}^{39}c_2 + {}^{13}c_4 \times {}^{39}c_1 + {}^{13}c_5}$

3) $\frac{{}^{13}c_5}{{}^{13}c_3 + {}^{13}c_4 + {}^{13}c_5}$ 4) $\frac{{}^{13}c_5}{{}^{13}c_3 \times {}^{13}c_4 \times {}^{13}c_5}$

31. If five cards are drawn at random from a pack of cards, the probability that they belong to different denominations is

1) $\frac{3223}{4165}$ 2) $\frac{2332}{4165}$ 3) $\frac{2112}{4165}$ 4) $\frac{1221}{4165}$

32. If 4 squares are selected at random on a chess board having 8×8 squares, then the probability that they will be in a diagonal line is

1) $\frac{4\sum_{n=4}^{8} {}^{n}C_4}{{}^{64}C_4}$ 2) $\frac{2\sum_{n=4}^{8} {}^{n}C_4}{{}^{64}C_4}$

3) $\frac{2\sum_{n=4}^{7} {}^{n}C_4 + {}^{8}C_4}{{}^{64}C_4}$ 4) $\frac{4\sum_{n=4}^{7} {}^{n}C_4 + 2\left({}^{8}C_4\right)}{{}^{64}C_4}$

33. Two numbers X and Y are chosen at random from the set {1, 2, ..., 3n}. The probability that $X^2 - Y^2$ is divisible by 3 is

1) $\frac{5n-3}{3(3n-1)}$ 2) $\frac{5n+3}{3(3n-1)}$

3) $\frac{5n-3}{3(3n+1)}$ 4) $\frac{5n}{3n-1}$

34. If A, B and C are three events such that $P(A)=0.3,\ P(B)=0.4,\ P(C)=0.8,$ $P(A\cap B)=0.12,\ P(A\cap C)=0.28,$ $P(A\cap B\cap C)=0.09$ and $P(A\cup B\cup C)\ge 0.75,$ then the limits of $P(B\cap C)$ are

1) $[0.32, 0.52]$ 2) $[0.23, 0.48]$

3) $[0.19, 0.44]$ 4) $[0.21, 0.19]$

35. A and B are two candidates seeking admission in IIT. The probability that A is selected is 0.5 and the probability that both A and B are selected is atmost 0.3. The probability of B getting selected cannot exceed

1) 0.6 2) 0.7 3) 0.8 4) 0.9

36. It is given that the events A and B are such that $P(A)=\frac{1}{4}, P\left(\frac{A}{B}\right)=\frac{1}{2}$ and $P\left(\frac{B}{A}\right)=\frac{2}{3}$ then P(B) = (AIEEE 2008)

1) $\frac{1}{6}$ 2) $\frac{1}{3}$ 3) $\frac{2}{3}$ 4) $\frac{1}{2}$

37. If a leap year is having 53 Sundays then the probability that the leap year contains exactly 52 mondays is

1) $\frac{1}{7}$ 2) $\frac{1}{3}$ 3) $\frac{2}{7}$ 4) $\frac{1}{2}$

38. Two symmetrical dice are thrown. The probability that the sum of the numbers appearing is 11, if 5 appears on the 1st die

1) $\frac{1}{18}$ 2) $\frac{1}{6}$ 3) $\frac{1}{3}$ 4) $\frac{1}{9}$

39. A symmetrical die is thrown 3 times and the sum of points thrown is found to be 15. The chance that the 1st throw was a four is

1) $\frac{1}{3}$ 2) $\frac{1}{4}$ 3) $\frac{1}{5}$ 4) $\frac{1}{6}$

40. A couple has 2 children. The probability that both are boys, if it is known that elder child is a boy is

1) $\frac{2}{3}$ 2) $\frac{1}{3}$ 3) $\frac{1}{2}$ 4) $\frac{3}{4}$

41. A bag contains 6 white and 4 black balls. Two balls are drawn at random and one is found to be white. The probability that the other ball is also white is

1) $\frac{2}{13}$ 2) $\frac{5}{13}$ 3) $\frac{8}{13}$ 4) $\frac{9}{13}$

42. A and B are two independent events such that $P(\bar{A} \cap B) = \frac{8}{25}$ and $P(A \cap \bar{B}) = \frac{3}{25}$, then P(A)=

1) $\frac{1}{5}$ 2) $\frac{3}{5}$ 3) $\frac{1}{5}$ or $\frac{3}{5}$ 4) $\frac{1}{2}$

43. A monkey seated before a type writer with 26 keys on the key board denoting the English alphabet. Then the probability for that monkey to type the word 'SIR' is

1) $\frac{^{26}C_3}{(26)^3}$ 2) $\frac{1}{^{26}C_3}$ 3) $\left(\frac{1}{26}\right)^3$ 4) $\frac{3}{26}$

44. A die is loaded so that six turns up twice as often as one and three times as often as any other face.The probability of getting an even number on the die if the die is rolled once is.

1) $\frac{2}{17}$ 2) $\frac{6}{17}$ 3) $\frac{10}{17}$ 4) $\frac{15}{17}$

45. The Intermediate Board has to select an examiner from a list of 100 persons. 40 of them women and 60 men; 50 of them knowing Telugu and 50 are not; 75 of them are teachers and the remaining are not. The probability that the board selects a telugu knowing woman teacher is

1) $\frac{1}{20}$ 2) $\frac{3}{20}$ 3) $\frac{17}{20}$ 4) $\frac{2}{20}$

46. A, B and C toss a coin one after another. Who ever gets head 1st will win the game. If A starts the game the probability of A's winning is

1) $\frac{1}{7}$ 2) $\frac{2}{7}$ 3) $\frac{4}{7}$ 4) $\frac{3}{7}$

47. A fair coin is tossed repeatedly. If tail appears on the first four tosses then the probability of head appearing on fifth toss is

1) $\frac{1}{2}$ 2) $\frac{1}{32}$ 3) $\frac{31}{32}$ 4) $\frac{1}{5}$

48. In a shooting test, the probabilities for 3 persons A, B, C to hit the target are $\frac{1}{2}, \frac{2}{3}$ and $\frac{3}{4}$. If all of them simultaneously aim at the target, the probability for exactly two persons hitting the target is

1) $\frac{1}{384}$ 2) $\frac{1}{12}$ 3) $\frac{11}{24}$ 4) $\frac{13}{24}$

49. A and B alternately throw a pair of symmetrical dice. A wins if he throws 6 before B throws 7 and B wins if he throws 7 before A throws 6. If A begins, the probability of his winning is

1) $\frac{11}{36}$ 2) $\frac{30}{61}$ 3) $\frac{31}{61}$ 4) $\frac{1}{36}$

50. An unbiased die is tossed until a number greater than 4 appears. The probability that an even number of tosses is needed is

1) $\frac{1}{2}$ 2) $\frac{2}{5}$ 3) $\frac{1}{5}$ 4) $\frac{2}{3}$

51. The probability of India winning a test match against West Indies is $\frac{1}{2}$. Assuming independence from match to match, the probability that in a 5 match series. India's second win occurs at third test is

1) $\frac{1}{8}$ 2) $\frac{1}{4}$ 3) $\frac{1}{2}$ 4) $\frac{2}{3}$

52. A bag contains 2 white, 3 black and 4 green balls. Two balls are drawn one after another with replacement. The probability that 1st one is white and second one is black is

1) $\frac{5}{27}$ 2) $\frac{1}{9}$ 3) $\frac{2}{27}$ 4) $\frac{4}{27}$

53. A bag contains 3 white, 2 black and 4 red balls. Three balls are drawn one after another with replacement at random. The probability of drawing a white, a black and a red ball in succession in that order is

1) $\frac{1}{21}$ 2) $\frac{5}{21}$ 3) $\frac{8}{243}$ 4) $\frac{2}{243}$

54. There are 6 red and 5 black balls in a bag. Two balls are drawn at random one after another with replacement. The probability that both the balls drawn may be red is

1) $\frac{30}{121}$ 2) $\frac{25}{121}$ 3) $\frac{18}{121}$ 4) $\frac{36}{121}$

55. In a group of equal number of men and women, 10% of men and 45% of women are unemployed. The probability that a person selected at random from that group is employed

1) $\frac{11}{40}$ 2) $\frac{29}{40}$ 3) $\frac{18}{40}$ 4) $\frac{9}{40}$

56. An urn A contains 8 black and 5 white balls. A second urn B contains 6 black and 7 white balls. A blind folded persons is asked to draw a ball selecting one of the urns, the probability that the ball drawn is white is

1) $\frac{5}{13}$ 2) $\frac{6}{13}$ 3) $\frac{7}{13}$ 4) $\frac{9}{13}$

57. One bag contains 5 white and 3 black balls and an other contains 4 white, 5 red balls. Two balls are drawn from one of them choosing at random. The probability that they are of different colours is

1) $\frac{15}{56}$ 2) $\frac{5}{18}$ 3) $\frac{275}{504}$ 4) $\frac{275}{624}$

58. A fair coin is tossed if the result is a head, a pair of fair dice is rolled and the score is noted. If the result is a tail, a card from a well shuffled pack of eleven numbered 1,2,3,.......11 is picked and the number on the card is noted. Then the probability that the number noted is either 7 or 8 is

1) $\frac{45}{512}$ 2) $\frac{193}{792}$ 3) $\frac{193}{1024}$ 4) $\frac{243}{792}$

59. If a randomly chosen year has 53 Sundays then probability that it is a non-leap year.

1) $\frac{3}{5}$ 2) $\frac{2}{5}$ 3) $\frac{1}{5}$ 4) $\frac{4}{5}$

60. The chance that Doctor A will diagonise disease X correctly is 60%. The chance that a patient will die by his treatment after correct diagnosis is 40% and the chance of death after wrong diagnosis is 70%. A patient of Doctor A who had disease X died. The probability that his disease was diagonised correctly is

1) 5/13 2) 6/13 3) 2/13 4) 7/13

61. A bag contains 6 balls two balls are drawn and found them to be red. The probability that five balls in the bag are red

1) $\frac{5}{6}$ 2) $\frac{12}{17}$ 3) $\frac{1}{3}$ 4) $\frac{2}{7}$

62. A man is known to speak the truth 2 out of 3 times. He throws a die and reports that it is a six. The probability that it is actually a five is

1) $\frac{3}{8}$ 2) $\frac{1}{7}$ 3) $\frac{2}{7}$ 4) $\frac{4}{5}$

63. For k = 1,2,3 the box B_k contains k red balls and (k+1) white balls. Let $P(B_1)=1/2, P(B_2)=1/3$ and $P(B_3)=1/6$. A box is selected at random and a ball is drawn from it. If a red ball is drawn, then the probability that it has come from box B_2 is (Eamcet 2008)

1) $\frac{35}{78}$ 2) $\frac{14}{39}$ 3) $\frac{10}{13}$ 4) $\frac{12}{13}$

64. Box A contains 2 black and 3 red balls while box B contains 3 black and 4 red balls. Out of these two boxes one is selected at random and the probability of choosing box A is double that of box B. If a red ball is drawn from the selected box, then the probability that it has come from box B is

1) 21/41 2) 10/31 3) 12/31 4) 13/41

65. Let E be an event which is neither a certainty nor an impossibility. If probability is such that $P(E)=1+\lambda+\lambda^2$ and $P(\overline{E})=(1+\lambda)^2$ in terms of an unknown λ then $P(E)$ is

1) 1 2) 3 / 4 3) 1 / 4 4) 1 / 2

66. There are 10 pairs of shoes in a cupboard, out of which 4 shoes are picked out one by one randomly, the probability that there is at least one correct pair is

1) $\frac{99}{323}$ 2) $\frac{89}{323}$ 3) $\frac{79}{323}$ 4) $\frac{69}{323}$

67. If $\frac{1+3p}{3}, \frac{1-p}{4}$ and $\frac{1-2p}{2}$ are the probabilities of the three mutually exclusive events, then $p \in$

1) $[0,1]$ 2) $\left[0,\frac{1}{2}\right]$ 3) $\left[\frac{1}{3},1\right]$ 4) $\left[\frac{1}{3},\frac{1}{2}\right]$

68. There are 5 balls numbered 1 to 5 and 5 boxes numbered 1 to 5. The balls are kept in the boxes one in each box. The probability that exactly 2 balls are kept in the correspoinding numbered boxes and the remaining 3 balls in the wrong boxes, is

1) $\frac{1}{5}$ 2) $\frac{1}{6}$ 3) $\frac{1}{10}$ 4) $\frac{1}{12}$

69. A 2 x 2 matrix is formed with entries from the set {0, 1}. The probability it is singular is

1) $\frac{3}{16}$ 2) $\frac{1}{8}$ 3) $\frac{5}{8}$ 4) $\frac{5}{16}$

70. A determinant is chosen at random from the set of all determinants of order 2 with elements 0 or 1 only. The probability that the determinant is negative is

1) 3/16 2) 3/8 3) 5/8 4) 7/8

71. A determinant is chosen at random from the set of all determinants of order 2 with elements 0 or 1 only. The probability that the determinant is non-zero is

1) 3/16 2) 3/8 3) 5/8 4) 7/8

72. A letter is known to have come from CHENNAI, JAIPUR, NAINITAL, MUMBAI. On the post mark only two consecutive letters AI are legible. The probability that it came from Mumbai is

1) $\frac{39}{190}$ 2) $\frac{42}{149}$ 3) $\frac{39}{191}$ 4) $\frac{38}{191}$

73. Four friends put their car keys on a table. when they leave the place, they picked up their keys at random. The probability that no persons picks his own key is

1) $\frac{3}{8}$ 2) $\frac{1}{4}$ 3) $\frac{5}{8}$ 4) $\frac{3}{4}$

74. If 5 cards letters as A, B, E, L and T shuffled placed one by one at random then probability of getting a word TABLE approximately is

1) 0.008 2) 0.8 3) 0.08 4) 0.0008

75. If A and B are two independent events such that $P(A)=\frac{1}{3}$ and $P(B)=\frac{3}{4}$, then $P\left\{\frac{B}{A\cup B}\right\}=$

1) $\frac{7}{10}$ 2) $\frac{8}{10}$ 3) $\frac{9}{10}$ 4) $\frac{6}{10}$

76. A bag contains 6 white balls and 4 black balls. A ball is drawn and is put back in the bag with 5 balls of the same colour as that of the ball drawn. A ball is drawn again at random. The probability that the ball drawn now is white is

1) $\frac{1}{5}$ 2) $\frac{2}{5}$ 3) $\frac{3}{5}$ 4) $\frac{4}{5}$

77. Bag *A* contains 4 white and 7 black balls. Bag *B* contains 5 white and 6 black balls. A die is rolled. If 2 or 5 turns up then choose bag *A* otherwise choose bag *B*. If one ball is drawn at random from the selected bag, then the probability that it is black is

1) $\frac{6}{11}$ 2) $\frac{7}{11}$ 3) $\frac{4}{11}$ 4) $\frac{19}{33}$

78. A survey shows that in a certain village 2 out of every 100 men and 1 out of every 100 women have stomach ulcers. A person selected at random from the village is found to have stomach ulcer. The probability that the person is a male, given that the probability of selecting a male from the village is 0.55 is

1) $\frac{22}{31}$ 2) $\frac{1}{2}$ 3) $\frac{1}{3}$ 4) $\frac{11}{31}$

79. A bag contains 5 balls the colours of which are known. Two balls are drawn and found them to be red. The probability that all the balls in the bag are red is

1) $\frac{1}{2}$ 2) $\frac{1}{3}$ 3) $\frac{1}{4}$ 4) $\frac{1}{5}$

80. A card from pack of 52 cards is lost. From the remaining cards of pack, two cards are drawn and are found to be spades. The probability of the missing card to be a spade is

1) $\frac{1}{50}$ 2) $\frac{3}{50}$ 3) $\frac{6}{50}$ 4) $\frac{11}{50}$

LEVEL-II (H.W.) - KEY

1) 3	2) 3	3) 2	4) 3	5) 1	6) 3	7) 3
8) 3	9) 1	10) 3	11) 3	12) 1	13) 1	14) 3
15) 2	16) 1	17) 1	18) 2	19)1	20) 2	21) 2
22) 2	23) 3	24) 3	25) 2	26) 2	27) 3	28) 2
29) 2	30) 2	31) 3	32) 4	33) 1	34) 3	35) 3
36) 2	37) 4	38) 2	39) 3	40) 3	41) 2	42) 3
43) 3	44) 3	45) 2	46) 3	47) 1	48) 3	49) 2
50) 2	51) 2	52) 3	53) 3	54) 4	55) 2	56) 2
57) 3	58) 2	59) 1	60) 2	61) 4	62) 2	63) 2
64) 2	65) 2	66) 1	67) 4	68) 2	69) 3	70) 1
71) 2	72) 2	73) 1	74)1	75) 3	76) 3	77) 4
78) 1	79)1	80) 4				

LEVEL-II (H.W.) - HINTS

1. $P(E)+P(\bar{E})=1, P(E).P(\bar{E})$ is maximum if $P(E)=P(\bar{E})$, max.of $P(E).P(\bar{E})=\frac{1}{4}$

2. Required probability $=\frac{{}^6c_1\times{}^6c_2+{}^6c_3}{{}^{12}c_3}=\frac{1}{2}$

3. $n(S)=15!$ $\quad n(E)=9!\times{}^{10}P_6$

4. $n(S)={}^{100}c_1$ $\quad n(E)=10$

5. $n(S)=3^5$ $\quad n(A)={}^5C_2.2^3$

6. $n(S)={}^{10}c_4$ $\quad n(\bar{E})={}^5c_2.{}^5c_2+{}^5c_4$

7. $n(S)={}^{80}c_2 \quad n(E)={}^{20}c_2$

8. $n(S)=8! \quad n(E)={}^8c_3.5!$

9. $n(S)={}^{25}c_2 \quad n(E)={}^7c_1.{}^4c_1+11=39$

10. $n(S)=6! \quad n(E)=\frac{6!}{4!}=30$

11. $n(S)=4^7 \quad n(E)=3^7$

12. $n(S)=7! \quad n(E)=5!\,2!$

13. $n(S)=20\times19 \quad n(E)=20(2)$

14. Req. Prob $=\frac{{}^4P_2}{4!}\left[1-\frac{1}{1!}+\frac{1}{2!}\right]$

15. $n(S)=3\times2\times1=6$; $\Delta\geq0$, $b=3$, $a=1$ or 2 $c=1$ or 3
$n(A)=2$

16. $R.P=\frac{4}{20}$

17. $n(S)=n^n$, $n(E)=\lfloor n$

18. $n(E)={}^{20}c_2$, $n(S)={}^{40}c_4$

19. $P(A)=\frac{{}^{10}C_4\times{}^6C_0\times2^0}{{}^{20}C_8}$

20. $n(S)={}^{10}c_3$; $E=\{(10,5,2)(8,2,4)(6,2,3)\}$, $n(E)=3$
$P(E)=\frac{1}{40}$

21. $n(E)=$ area of the triangle ABE
$=\frac{1}{2}\times3\times4=6sq.units$
$n(s)=$ Area of the rectangle ABCD
$=4\times15=60\,sq.units \quad \therefore P(E)=\frac{6}{60}=\frac{1}{10}$

22. $p=\frac{210}{n(s)}, q=\frac{4}{n(s)}, r=\frac{12\times7\times4}{n(s)} \qquad q<p<r$

23. Req. Probability $=\frac{6}{36}=\frac{1}{6}$

24. P(sum is even) = P(even) P(even) + P(odd no)
P(odd no) $=3\frac{1}{9}\times3\frac{1}{9}+3\frac{2}{9}\times3\times\frac{2}{9}=\frac{1}{9}+\frac{4}{9}=\frac{5}{9}$

25. $n(s)=6^4$ $\quad A=\{(1\ 1\ 1\ 3),(1\ 1\ 2\ 2)\}$
$n(A)=\frac{4!}{3!}+\frac{4!}{2!2!}=10$

26. $p=\frac{1}{36}, q=\frac{5}{36} \qquad p:q=1:5$

27. B be the event that the leap year has 53 Fridays
A be the event that leap year has 53 Thursday.
$P\left(\frac{A}{B}\right)=\frac{1}{2}$

28. $P(E)=\frac{{}^4c_1\times{}^{48}c_{12}}{{}^{52}c_{13}}$

29. $P(E)=\frac{{}^2c_2\times{}^{37}c_{11}}{{}^{39}c_{13}}$

30. Req. $=\frac{{}^{13}c_5}{{}^{13}c_3\times{}^{39}c_2+{}^{13}c_4\times{}^{39}c_1+{}^{13}c_5}$

31. Req. $=\frac{{}^{13}c_5\left({}^4c_1\right)^5}{{}^{52}c_5}$

32. $n(E)=2\left({}^8c_4+2\left({}^7c_4+{}^6c_4+{}^5c_4+{}^4c_4\right)\right)$

33. Here random experiment is selection of two numbers X and Y from the set $\{1, 2, 3, ..., 3n\}$
Let S = the simple space and E = the event that $X^2 - Y^2$ is divisible by 3.
$X^2 - Y^2$ will be divisible by 3 if $(X - Y)(X + Y)$ is divisible by 3. i.e., either $X - Y$ or $x + y$ must be divisible by 3
Let us arrange the consecutive numbers starting from 1, say, in three columns as follows :

I	II	III
1	2	3
4	5	6
7	8	9
$3n-2$	$3n-1$	$3n$

where each column contains n numbers. Select two numbers either from I or II or III then either the sum or difference is divisible by 3. Select one number from I and one from II. Then only the sum is divisible by 3.

$\therefore n(E) = 3\,{}^nC_2 + {}^nC_1 \cdot {}^nC_1 = \frac{3n(n-1)}{2} + n^2$

$= \frac{n}{2}[3(n-1)+2n] = \frac{n}{2}(5n-3)$

$\therefore$ Required probability $P(E) = \frac{n(E)}{n(S)} = \frac{5n-3}{3(3n-1)}$

34. $P(A\cup B\cup C) \le 1$, given $P(A\cup B\cup C) \le 0.75$

$\Rightarrow P(B\cap C) \in [0.19, 0.44]$

35. Given P(A) = 0.5 and $P(A\cap B) \le 0.3$, we have
P(A) + P(B) − $P(A\cup B)$ $= P(A\cup B) \le 1$

$\Rightarrow P(A)+P(B) \le 1 + P(A\cup B) \le 1+0.3 = 1.3$

$\Rightarrow 0.5 + P(B) \le 1.3 \Rightarrow P(B) \le 0.8$

Thus probability of B getting selected cannot exceed 0.8

36. $\frac{P(A\cap B)}{P(B)} = \frac{1}{2}, \frac{P(A\cap B)}{P(A)} = \frac{2}{3}$

Hence $\frac{P(A)}{P(B)} = \frac{3}{4}$ (But $P(A) = \frac{1}{4}$) $\Rightarrow P(B) = \frac{1}{3}$

37. S= { (sat, Sun), (Sun, Mon) } E= { (Sat, Sun) }

38. $n(A) = 6, n(A\cap B) = 1, p\left(\frac{B}{A}\right) = \frac{1}{6}$

39. $p\left(\frac{B}{A}\right) = \frac{p(A\cap B)}{p(A)} = \frac{2}{10} = \frac{1}{5}$

40. $P(E) = \frac{1}{2}$ $\qquad n(S) =$ BG, BB

41. R.P is $\dfrac{\frac{{}^6c_2}{{}^{10}c_2}}{\frac{{}^6c_1 \cdot {}^4c_1}{{}^{10}c_2} + \frac{{}^6c_2}{{}^{10}c_2}} = \frac{5}{13}$

42. $P(\overline{A})P(B) = \frac{8}{25}$; $P(A)P(\overline{B}) = \frac{3}{25} \therefore P(A) = \frac{1}{5}$ or $\frac{3}{5}$

43. $P(E) = \frac{1}{26}.\frac{1}{26}.\frac{1}{26} = \left(\frac{1}{26}\right)^3$

44. Let A_K be the event that the die shows the number k on the top face for $k = 1, 2, 3, 4, 5, 6$.

Given, $P(A_6) = 2P(A_1)$
$P(A_6) = 3P(A_2) = 3P(A_3) = 3P(A_4) = 3P(A_5)$
Here $A_1, A_2, A_3, A_4, A_5, A_6$ are exclusive and exhaustive $\therefore \sum_{k=1}^{6} P(A_k) = 1$ By data,
$2P(A_1) = 3P(A_2) = 3P(A_3) = 3P(A_4) = 3P(A_5) =$

$P(A_6) \Rightarrow \frac{P(A_1)}{3} = \frac{P(A_2)}{2} = \frac{P(A_3)}{2} = \frac{P(A_4)}{2} = \frac{P(A_5)}{2}$

$= \frac{P(A_6)}{6} = \frac{\sum_{k=1}^{6} P(A_k)}{17} = \frac{1}{17}$

$\therefore P(A_1) = \frac{3}{17}, P(A_2) = P(A_3) = P(A_4) = P(A_5) = \frac{2}{17},$

$P(A_6) = \frac{6}{17}$

P(getting an even number) $= P(A_2 \cup A_4 \cup A_6)$

$= P(A_2) + P(A_4) + P(A_6) = \frac{2}{17} + \frac{2}{17} + \frac{6}{17} = \frac{10}{17}$

45. $P(W) = \frac{40}{100}, P(M) = \frac{60}{100}; P(T_k) = \frac{50}{100}, P(T_{uk}) = \frac{50}{100}$

$P(\text{Teach}) = \frac{75}{100}, P(NT) = \frac{25}{100}; P(T\cap W\cap \text{Teach}) = \frac{3}{20}$

46. $P(E) = \frac{p}{1-q^3}$

47. $P(E) = \frac{1}{2}$

48. A, B, C be the events- Req. Prob

$= P(A\cap B\cap \overline{C}) + P(A\cap \overline{B}\cap C) + P(\overline{A}\cap B\cap C) = \frac{11}{24}$

49. a = P (A getting 6) ; b = P (B getting 7)

$\lambda = (1-a)(1-b)\,p(A) = \frac{a}{1-\lambda} = \frac{\frac{5}{36}}{1-\frac{31}{36}\frac{5}{6}} = \frac{30}{61}$

50. $p = \frac{1}{3}, q = \frac{2}{3}$; Req. Prob $= \frac{pq}{1-q^2} = \frac{2}{5}$

51. W L W $\frac{W}{L}\frac{W}{L}$ or LWW $\frac{W}{L}\frac{W}{L} = \frac{1}{2}\times\frac{1}{2}\times\frac{1}{2} + \frac{1}{2}\times\frac{1}{2}\times\frac{1}{2}$

52. Req. Prob $= \frac{2}{9}\times\frac{3}{9} = \frac{2}{27}$

53. Req. Prob $= \frac{3}{9}\times\frac{2}{9}\times\frac{4}{9} = \frac{8}{243}$

54. Req. Prob $= \frac{6}{11}\times\frac{6}{11} = \frac{36}{121}$

55. $P(E) = P(B_1).P\left(\frac{A}{B_1}\right) + P(B_2).P\left(\frac{A}{B_2}\right)$

$= \frac{1}{2}.\frac{90}{100} + \frac{1}{2}.\frac{55}{100} = \frac{29}{40}$; B_1 = selection of man,
B_2 = selection of woman; A= employed

56. R.P is $\frac{1}{2}\left[\frac{5}{13}+\frac{7}{13}\right]$

57. R.P is $\frac{1}{2}\left[\frac{{}^5c_1\times{}^3c_1}{{}^8c_2}+\frac{{}^4c_1\times{}^5c_1}{{}^9c_2}\right]$

58. $P(E) = P(E_1).P\left(\frac{E}{E_1}\right)+P(E_2).P\left(\frac{E}{E_2}\right)$

E_1 = getting head $\qquad E_2$ = getting tail

$=\frac{1}{2}.\frac{11}{36}+\frac{1}{2}.\frac{2}{11}=\frac{1}{2}\left(\frac{11}{36}+\frac{2}{11}\right)=\frac{193}{792}$

59. $P(E)=\frac{P(NL).P\left(\frac{S_{53}}{NL}\right)}{P(L).P\left(\frac{S_{53}}{L}\right)+P(NL).P\left(\frac{S_{53}}{NL}\right)}=\frac{\frac{3}{4}.\frac{1}{7}}{\frac{1}{4}.\frac{2}{7}+\frac{3}{4}.\frac{1}{7}}=\frac{3}{5}$

60. Req.prob = $\frac{\frac{60}{100}.\frac{40}{100}}{\frac{60}{100}.\frac{40}{100}+\frac{40}{100}.\frac{70}{100}}$

61. There may be 2 red or 3 or 4 or 5 or 6 red balls in the bag

Req. prob = $\frac{\frac{1}{5}\left(\frac{{}^5C_2}{{}^6C_2}\right)}{\frac{1}{5}\left(\frac{{}^2C_2}{{}^6C_2}+\frac{{}^3C_2}{{}^6C_2}+\frac{{}^4C_2}{{}^6C_2}+\frac{{}^5C_2}{{}^6C_2}+\frac{{}^6C_2}{{}^6C_2}\right)}$

62. $\text{Req.}=\frac{\frac{1}{3}\times\frac{1}{6}}{\frac{2}{3}\times\frac{1}{6}+\frac{5}{6}\times\frac{1}{3}}$

63. $\text{R.P}=\frac{\frac{1}{3}.\frac{2}{5}}{\frac{1}{2}.\frac{1}{3}+\frac{1}{3}.\frac{2}{5}+\frac{1}{6}.\frac{3}{7}}$

64. P(A) = 2 P(B) $\Rightarrow$ P(B) = $\frac{1}{3}$ P(A) = $\frac{2}{3}$

R.P = $\frac{\frac{1}{3}.\frac{4}{7}}{\frac{2}{3}.\frac{3}{5}+\frac{1}{3}.\frac{4}{7}}$

65. $P(E)+P(\overline{E})=1\Rightarrow\lambda=-1,\ -1/2$

If $\lambda=-1$ then $P(E)=1$ (not possible)

66. $P(E)=\frac{20\times18\times16\times14}{20\times19\times18\times17}=\frac{224}{323}$

$1-P(E)=1-\frac{224}{323}=\frac{99}{323}$

67. $\frac{1+3p}{3}\ge0,\frac{1-p}{4}\ge0,\frac{1-2p}{2}\ge0$

$\Rightarrow p\ge-\frac{1}{3},\ p\le1,\ p\le\frac{1}{2}$

$\frac{1+3p}{3}+\frac{1-p}{4}+\frac{1-2p}{2}\le1\Rightarrow p\ge\frac{1}{3}\quad\therefore\frac{1}{3}\le p\le\frac{1}{2}.$

68. $n(E)=5_{P_3}\left[\frac{1}{2!}-\frac{1}{3!}\right],\ n(s)=120$

69. Out of 16 matrices six are non singular

$\begin{pmatrix}1&1\\0&1\end{pmatrix},\begin{pmatrix}1&1\\1&0\end{pmatrix},\begin{pmatrix}0&1\\1&1\end{pmatrix},\begin{pmatrix}1&0\\1&1\end{pmatrix},\begin{pmatrix}1&0\\0&1\end{pmatrix},\begin{pmatrix}0&1\\1&0\end{pmatrix}$

P(E) = $1-\frac{6}{16}=\frac{5}{8}$

70. $n(s)=2^4,n(E)=3$

71. $n(s)=2^4,n(E)=6$

72. P(E) = $\frac{\frac{1}{5}}{\frac{1}{6}+\frac{1}{5}+\frac{1}{7}+\frac{1}{5}}=\frac{42}{149}$

73. The number of derangements of 4 distinct elements

is $\lfloor 4\left(1-\frac{1}{\lfloor 1}+\frac{1}{\lfloor 2}-\frac{1}{\lfloor 3}+\frac{1}{\lfloor 4}\right)=12-4+1=9$

P(E= $\frac{9}{\lfloor 4}=\frac{9}{24}=\frac{3}{8}$

74. $\frac{1}{5}\times\frac{1}{4}\times\frac{1}{3}\times\frac{1}{2}=0.008$

75. $P\left(\frac{B}{A\cup B}\right)=\frac{P(B)}{P(A\cup B)}$

76. $P(E)=\frac{11}{15}\times\frac{3}{5}+\frac{6}{15}\times\frac{2}{5}=\frac{3}{5}$

77. $P(E)=P(A_1)P(E/A_1)+P(A_2)P(E/A_2)$

$=\frac{2}{6}\times\frac{7}{11}+\frac{4}{6}\times\frac{6}{11}\quad=\frac{38}{66}=\frac{19}{33}$

78. Req. probability $=\frac{0.55\times\frac{2}{100}}{0.55\times\frac{2}{100}+0.45\times\frac{1}{100}}=\frac{110}{155}=\frac{22}{31}$

79. Required probability $\frac{1}{\frac{1}{10}+\frac{3}{10}+\frac{6}{10}+\frac{10}{10}}=\frac{1}{2}$

80. Required probability $=\frac{{}^{12}C_2}{{}^{12}C_2+3({}^{13}C_2)}=\frac{11}{50}$